

Traps, open items and things that fail silently

Ordered by how expensive the mistake is. 'Fails silently' means the tools report success and the symptom appears somewhere else entirely.

Area	Item	Detail
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Fails silently - highest cost first

Area	Item	Detail
Build	LUT_DIR macro	Locates openofdm's three ROM .mif files for \$readmemh. Both synthesis and XSim resolve a relative \$readmem path against the TOOL's working directory, not the source file's, so it cannot have a relative default. Set by [openofdm::defines].
Build	OWIFI_SRC macro	Locates font8x16.mem for text_screen. Same mechanism, set by build_rasbb.tcl.
Build	Check the three \$readmem lines	After 'make check' or a synth run, confirm the log contains three 'INFO: [Synth 8-3876] ... is read successfully' lines for rot_lut, atan_lut and deinter_lut_from_coe.
Flash	Power-cycle before every flash write	WP#/IO2 on IC5 and IC7 is unconnected - confirmed FROM PCB (pad 3 unconnected on both). A warm write silently does nothing.
Vivado	.xpr element order	Inside a <Config> block every <Option Name="VerilogDir"> (the include path) must come BEFORE any <Verilog_Define>. Put a define first and Vivado drops the entire include path on load - no error, no warning.
Simulation	Command-line -d beats an `include-d `define	Measured precedence, the opposite of the intuition. A stale verilog_define persisted in a .xpr once produced three consecutive plausible-looking runs against an old vector.
Simulation	Packed-hex vectors must be converted	The bench reads two decimal columns per line (I Q). Most captures under testing_inputs/conducted/ are packed hex.

Hardware - open or needing verification

Area	Item	Detail
Constraints	Bank 14 VCCO is +3V3 but the debug ports are LVCMOS18	RASBB.kicad_pcb has VCCO_14 on +3V3 (6 pins), and J5/J6 carry +3V3 on pins 2 and 20. rasbb_rasrf.xdc constrains dbgBitClk, dbgFrameRdy, dbgIser, dbgQser, dbgOutI[*], dbgOutQ[*], Key0 and Key1 as LVCMOS18. This is the SAME class of mismatch as the bank-15 SPI/TMDS pins, which were corrected on 2026-08-02 after the PCB was checked.
RASRF	U8 part number not specified	The clock squarer between the 40 MHz TCXO and CLK_REF is a SOT-23-5 whose schematic Value is '~'.
RASRF	U8 inputs flipped	Findings_RevA.md item 3: U8's + and - inputs are exchanged; the fix is to swap pads 3 and 4.
HDMI	Phantom power through the TMDS pins	A connected monitor's TMDS terminations back-feed +3V3 through FPGA bank-15 clamp diodes; the TPD12S016 does not block TMDS. Planned rework: 8 x 100 nF in series between the IC11 tap and J3, plus 8 x 270 R bias.
Front end	External 100 R on every used LVDS pair	The ADC pins are DIFF_SSTL18_II in an HR bank at VCCO 1.8 V. DIFF_TERM is inert there (LVDS_25 / 2.5 V only, UG471), and IN_TERM's Thevenin split to 0.9 V fights the ADC's ~1.2 V common mode.
Front end	Reconfigure the FPGA after any ADC re-init	After any reconfiguration or reset of the ADC - for example an STM32 firmware reflash that re-runs ADC init - the link can slip word alignment in a way the FCLK watchdog cannot see.

Design rules that must not be relaxed

Area	Item	Detail
Timing	DCLK is a TRUE 120 MHz clock	Constrain it at 8.333 ns. The old 16.667 ns constraint hid real failures that came and went with placement luck - RASM2400 'finding D1'.
Timing	Setup is not negotiable, hold on the ADC pads is	build_rasbb.tcl refuses to write a bitstream when WNS < 0. Hold is false-pathed on ADC_chA/B/C only, because the runtime IDELAY calibration re-centres the eye and the static min-delay number says nothing about it. Setup stays analysed everywhere.
Timing	The Viterbi ACS is the critical path	~15 logic levels, ~60% route delay, and congestion-sensitive: adding the frame buffer's BRAMs alone once cost 175 ps and pushed it negative with default directives.
Front end	Word assembly order	{lane_cur, lane_hist} - the NEWER word in the HIGH half. Do not 'fix' it back to {hist, cur}.
Pins	Derive pins from the .kicad_pcb	RASBB has a known .net-vs-.kicad_pcb desync. A stale RASM2400 constraint set mapped TMDS onto D4/C4/E1/D1/F2/E2/G2 - PCIe RX lanes, GND and +1V8 on this board - and would have collided with ADC_chA_N and ADC_chB_N.
Vivado	Never run upgrade_ip -quiet	The openofdm .xci were saved for whatever part they were last used on. upgrade_ip retargets them; a silent failure leaves the IP locked.
Vivado	Do not build OWIFI from a GUI project	The batch flow (tools/build_rasbb.tcl) is authoritative - it produced every bitstream that ran on hardware. owifi.xpr and RASBB/Software/AP_FPGA/AP_FPGA.xpr carry absolute paths and are not portable.
Simulation	Regenerate simulation targets after a bitstream build	build_rasbb.tcl does reset_target all and then regenerates only the SYNTHESIS target, so simulation products go missing.

Receiver limits - functional, not defects

Area	Item	Detail
Receiver	54 Mbps and MCS7 are margin-limited	64-QAM 3/4 and 64-QAM 5/6 are the least-protected rates in the set. A 14-byte 54 Mbps frame decodes; a 1537-byte one does not. MCS7 passes 3 of 9.

Receiver	802.11b is not supported
Receiver	5 GHz / real 802.11a is not reachable
Receiver	HT Greenfield untested
Frame log	CFO sign convention uncalibrated
Front end	ramp_checker over-counts
Tuning	CR_POWER_THRES wants a re-sweep

DSSS/CCK is an entirely different PHY - Barker spreading and CCK correlation. No such logic exists anywhere in the tree.

The MAX2831 is a 2.4 GHz transceiver. The PHY itself already decodes all 11a rates.
The flow detects L-SIG and then HT-SIG, i.e. HT mixed mode. Greenfield frames carry no L-SIG, and no vector was available.
The magnitude is measured; which oscillator is high has never been checked against a generator. The sign is kept even when the magnitude truncates to zero (' -0K') so a calibration has something to read.
Its BER figure over-counts, per the RASPMO work.
The power_trigger threshold has not been re-swept on this hardware.

What happens if you get it wrong	Status / source
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Synthesis only WARNS, the ROMs load as X, and the receiver decodes nothing while looking perfectly healthy.	known, documented
The HDMI frame log comes up garbled. Synthesis only warns.	known, documented
Without this check a broken build is indistinguishable from a working one until nothing decodes.	procedure
You flash, nothing changes, and you debug the wrong thing.	hardware fact
Synthesis cannot open common_params.v, and the cause is nowhere near the symptom. Set both through tcl instead, so Vivado writes canonical XML by construction.	known, documented
A simulation 'passes' against the wrong capture. Every sim script now rewrites the define list explicitly, and the bench writes the macro it actually compiled into sample_file_name.txt - check that file, not your intent.	mitigated
A vector feeds in and nothing happens. Convert with scripts/conv_iq_hex.py first.	known

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What happens if you get it wrong	Status / source
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The bank is internally consistent to Vivado (every bank-14 port uses the same standard), so nothing is flagged - but the pins actually swing 3.3 V, and the declared drive/slew numbers are those of a standard the bank is not running. The inputs (Key0/Key1) still read correctly. Worth deciding deliberately rather than by inheritance.	OPEN - verify VCCO_14 on the assembled board, then either change to LVCMOS33 or record why not
The BOM cannot be built as-is.	OPEN - RASRF2400WB Rev A
The reference squarer does not work as drawn.	OPEN - Findings_RevA.md
The board is partly powered through its HDMI output. Also: once the AC-coupling rework goes in, a push-pull DIFF_SSTL18 drive would exceed the sink's absolute maximum - which is one more reason TMDS_33 must stay.	known, rework planned
Without the external termination the eye closes and the link behaves like an RF problem.	hardware fact
Silent, persistent misalignment. Reconfigure the FPGA or toggle i_rst afterwards.	hardware fact

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Intermittent decode garbage that looks like an RF problem.	rule
Shipping a setup violation means shipping intermittent, RF-looking decode errors.	rule
Anything added to the design can break timing from a distance. Re-check WNS after every addition - the HDMI branch in particular.	rule
That ordering was a MEASURED 1-error-per-16-words assembly bug: 2.5 M errors/s at 20 MSPS, flat across all IDELAY taps.	rule
Pin collisions with the ADC inputs, found late.	rule
Synthesis dies on a missing module while the IP reports 'up to date'.	rule
A GUI build can differ from the shipped one in ways nobody tracks.	rule
'Can not find file: ../div_gen_xlslice/sim/div_gen_xlslice_0_0.v'. Run generate_target simulation first.	known

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Not a demodulation fault: the count of frames containing the exact expected MAC pair equals the FCS-OK count at every MCS, so a frame either comes out perfect or carries a handful of wrong bits.	MEASURED 2026-08-01
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Adding it means a second receive chain plus arbitration with the OFDM path - a project, not a change.

Needs a different front end, not firmware.

Flagged rather than asserted - probably unsupported.

Do not trust the direction until it is calibrated once.

Treat it as a go/no-go indicator, not a measurement.

Detection sensitivity is not at its optimum.

by design

by design

IN RTL, UNTESTED

IN RTL, UNTESTED

known

OPEN